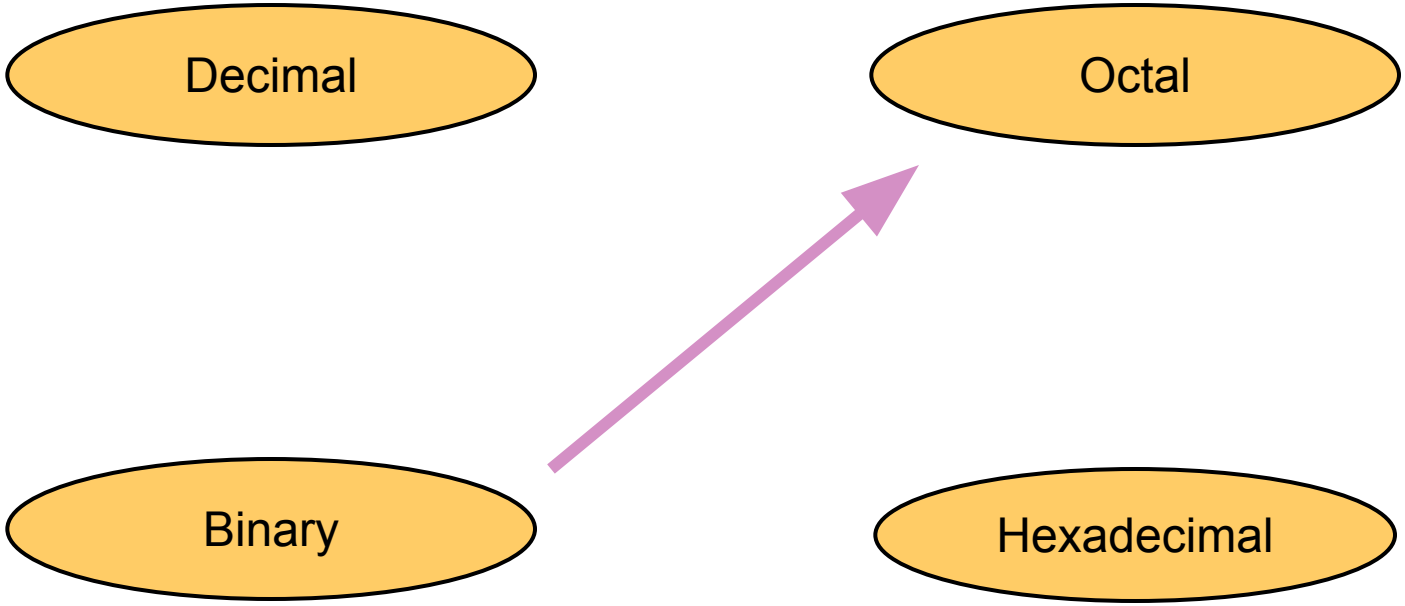
Binary to Octal



Binary to Octal

- Technique
  - Group bits in threes, starting on right
  - Convert to octal digits

Example

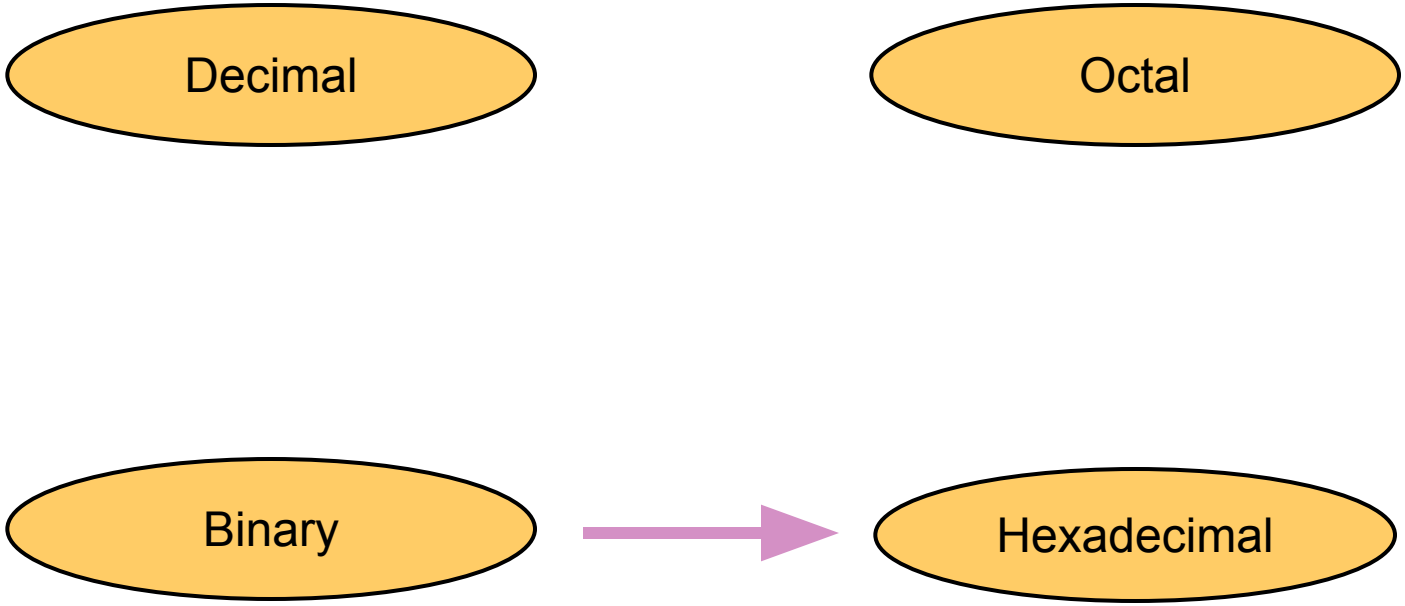
$1011010111_2 = ?_8$

1 011 010 111

1 3 2 7

$1011010111_2 = 1327_8$

Binary to Hexadecimal



Binary to Hexadecimal

- Technique
  - Group bits in fours, starting on right
  - Convert to hexadecimal digits

Example

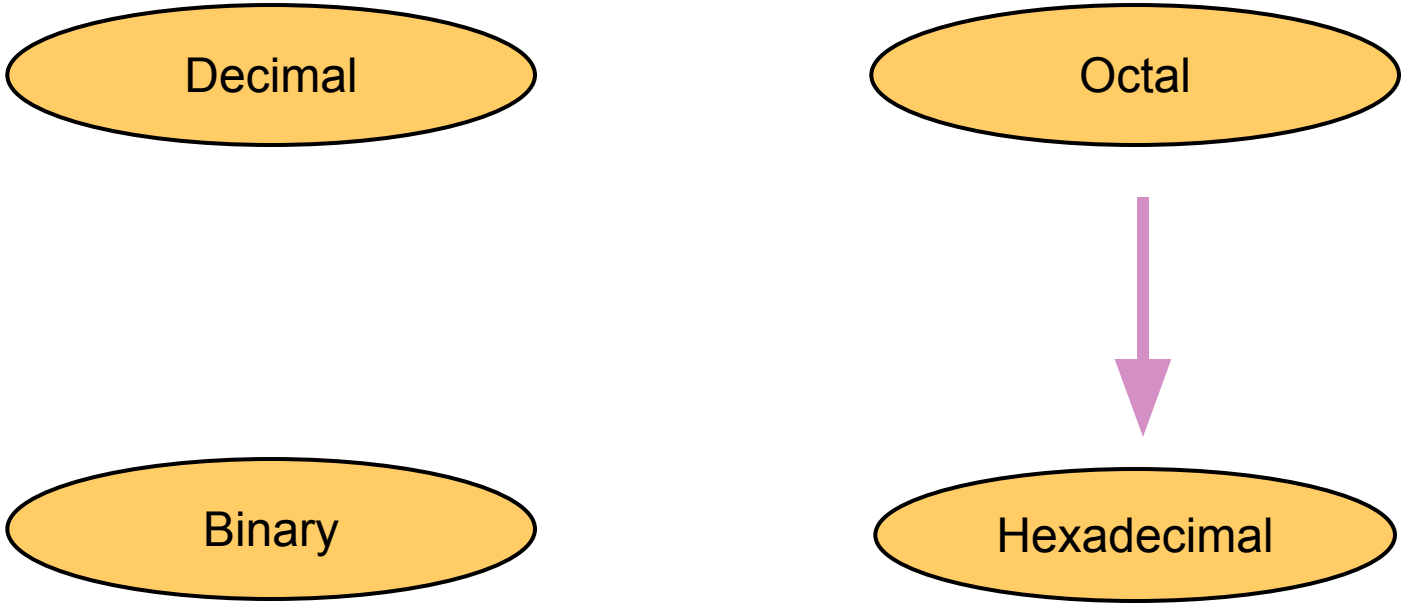
$1010111011_2 = ?_{16}$

10 1011 1011

2 B B

$1010111011_2 = 2BB_{16}$

Octal to Hexadecimal



Octal to Hexadecimal

- Technique
  - Use binary as an intermediary

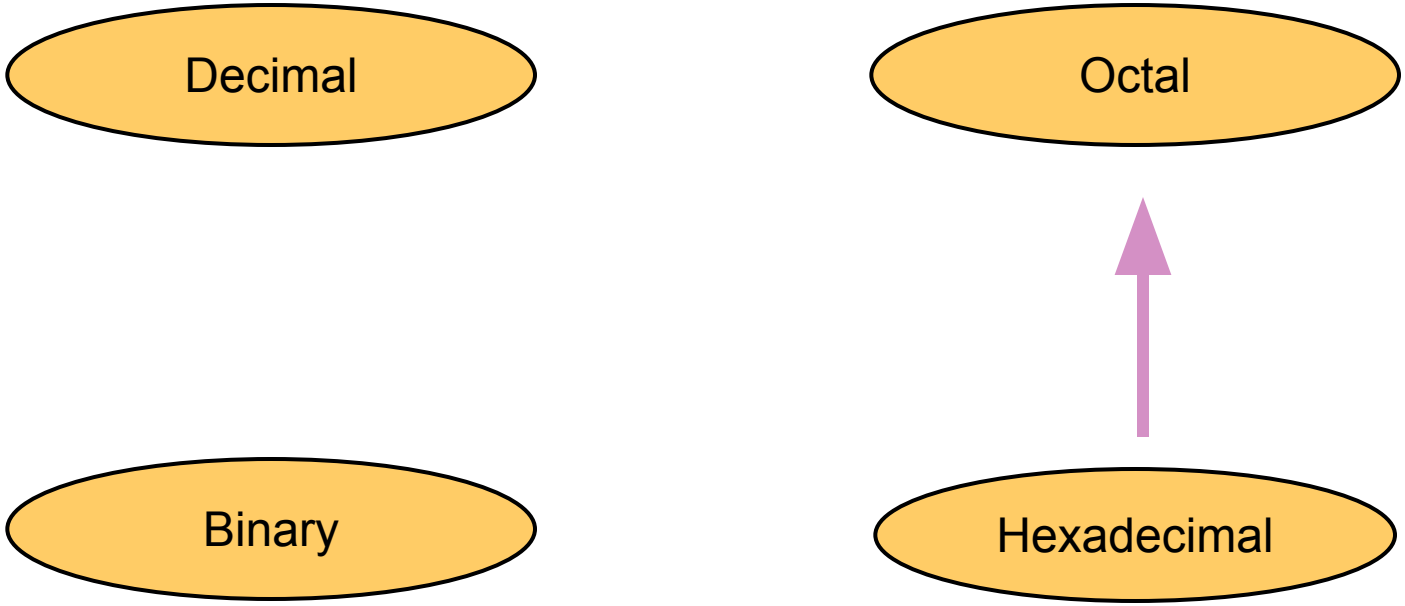
Example

$1076_8 = ?_{16}$

1	0	7	6
001	000	111	110
2	3		E

$1076_8 = 23E_{16}$

Hexadecimal to Octal



Hexadecimal to Octal

- Technique
  - Use binary as an intermediary

Example

$1F0C_{16} = ?_8$

1	F	0	C
0001	1111	0000	1100
1	7	4	1	4

$1F0C_{16} = 17414_8$

Exercise – Convert ...

Decimal	Binary	Octal	Hexa-decimal
43			
	111011		
		503	
			1BF

Common Powers

Base 10

Power	Preface	Symbol	Value
10^{-12}	pico	p	.000000000001
10^{-9}	nano	n	.000000001
10^{-6}	micro	μ	.000001
10^{-3}	milli	m	.001
10^{3}	kilo	k	1000
10^{6}	mega	M	1000000
10^{9}	giga	G	1000000000
10^{12}	tera	T	1000000000000

Common Powers

Base 2

Power	Preface	Symbol	Value
2^{10}	kilo	k	1024
2^{20}	mega	M	1048576
2^{30}	Giga	G	1073741824

Fractions

Decimal to binary

3.14579

11.001001...

```
 .14579
x      2
0.29158
x      2
0.58316
x      2
1.16632
x      2
0.33264
x      2
0.66528
x      2
1.33056
etc.
```

Fractions

Binary to decimal

$$
\begin{aligned}
10.1011 \Rightarrow \quad & 1 \times 2^{-4} = 0.0625 \\
& 1 \times 2^{-3} = 0.125 \\
& 0 \times 2^{-2} = 0.0 \\
& 1 \times 2^{-1} = 0.5 \\
& 0 \times 2^{0} = 0.0 \\
& 1 \times 2^{1} = 2.0 \\
\hline
& \quad 2.6875
\end{aligned}
$$

How to convert the hexadecimal number $(1A.23)_{16}$ to decimal

Hexadecimal	1		A	•	2		3
Place values	16^1		16^0		16^{-1}		16^{-2}
Partial result	16	+	10	+	0.125	+	0.012

Decimal: 26.137

Note that the result in the decimal notation is not exact, because $3 \times 16^{-2} = 0.01171875$. We have rounded this value to three digits (0.012).

How to convert $(23.17)_8$ to decimal.

Octal	2		3	•	1		7
Place values	8^1		8^0		8^{-1}		8^{-2}
Partial result	16	+	3	+	0.125	+	0.109

Decimal: 19.234

This means that (23.17)8 ≈ 19.234 in decimal. Again, we have rounded up $7 \times 8^{-2} = 0.109375$.